2025 VCE Specialist Mathematics Year 12 Trial Examination 2

Kilbaha Education

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Kilbaha Education (Est. 1978) (ABN 47 065 111 373)
PO Box 3229
Cotham Vic 3101
Australia
PayID: 47065111373
Email: kilbaha@gmail.com
Tel: +61 413 425 374
Web: https://kilbaha.com.au

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Write your **student number** in the boxes above. **Letter**

Specialist Mathematics Examination 2

Question and Answer Book VCE Examination

- Reading time is **15 minutes**:
- Writing time is **2 hours**:

Approved materials

- Protractors, set squares and aids for curve sketching.
- One bound reference
- One approved CAS calculator or CAS software, and one scientific calculator.

Materials supplied

- Question and Answer Book of 36 pages.
- Formula Sheet
- Multiple-Choice Answer Sheet.

Instructions

- Follow the instructions on your Multiple-Choice Answer Sheet.
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this book.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents pages

Section A (20 questions, 20 marks) _____ 4-10

Section B (6 questions, 60 marks) _____ 11-28

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Section A – Multiple-choice questions

Instructions

- Answer **all** questions in pencil on your Multiple-Choice Answer Sheet.
- Choose the response that is **correct** for the question.
- A correct answer scores 1; an incorrect answer scores 0.
- Mark will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the acceleration due to gravity to have a magnitude $g\ \text{m s}^{-2}$, where $g = 9.8$

Question 1

Consider the following statement.

If you eat your dinner, then you can have ice-cream or cake.

The contrapositive statement is

A. If you don't eat your dinner, then you cannot have ice-cream or cake.

B. If you don't eat your dinner, then you cannot have ice-cream and you can not have cake.

C. If you didn't have ice-cream and didn't have cake then you didn't eat your dinner.

D. If you didn't have ice-cream or cake then you didn't eat your dinner.

Question 2

Consider the function with the rule $f(x) = \dfrac{x^2 - a^2}{x - b}$, where $a \in R^+$ and $b \in R$.

Which of the following is **false**?

A. The graph of the function has two straight line asymptotes if $|b| \neq a$.

B. The graph of the function has a point of discontinuity if $|b| = a$.

C. The graph has two turning points if $|b| > a$.

D. The graph always has a point of inflection.

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Question 3

Given that $|z-a-2ai|=r$ where $0<r<a$, the maximum value of $|z|$ is

A. $\sqrt{5}a+r$

B. $\sqrt{5}a-r$

C. $\sqrt{2}a+r$

D. $\sqrt{2}a-r$

Question 4

Consider the line $L_1:\ \underset{\sim}{r}(t)=(3+2t)\underset{\sim}{i}+(2-t)\underset{\sim}{j}+(1-3t)\underset{\sim}{k},\quad t\in R,$ and the two planes

$\Pi_1:\ x+2y+3z=1$ and $\Pi_2:\ x+2y-z=3$, then

A. The line intersects both planes Π_1 and Π_2 at unique points.

B. The line intersects the plane Π_1 but does not intersect the plane Π_2.

C. The line intersects the plane Π_2 but does not intersect the plane Π_1.

D. The line is the intersection of the two planes Π_1 and Π_2.

Question 5

Given the vectors $\underset{\sim}{a}=4\underset{\sim}{i}+t\underset{\sim}{j}-5\underset{\sim}{k}$ and $\underset{\sim}{b}=-2\underset{\sim}{i}+\underset{\sim}{j}-t\underset{\sim}{k}$, if the scalar resolute of $\underset{\sim}{a}$ in the direction $\underset{\sim}{b}$, is twice the scalar resolute of $\underset{\sim}{b}$ in the direction $\underset{\sim}{a}$, then t is equal to

A. $\pm\sqrt{21}$

B. $-\sqrt{21}$ only.

C. $\dfrac{4}{3}$

D. $\pm\sqrt{7}$

Question 6

The equation $y=2z+3$ represents

A. a plane parallel to the x-axis.

B. a plane parallel to the YZ plane.

C. a line parallel to the YZ plane.

D. a line parallel to the x-axis

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Question 7

Let z be a complex number, in the Argand plane, $|z-a|=2|z-ai|$ where $a \in R\setminus\{0\}$ represents

A. a straight line

B. a circle

C. an ellipse

D. an hyperbola

Question 8

The distance of the plane $\underset{\sim}{r}(s,t)=4\underset{\sim}{i}+\beta\underset{\sim}{j}-\underset{\sim}{k}+t\left(3\underset{\sim}{i}+4\underset{\sim}{j}+2\underset{\sim}{k}\right)+s\left(\underset{\sim}{i}+3\underset{\sim}{j}-\underset{\sim}{k}\right)$, where $s,t \in R$, to the origin is $\sqrt{6}$ then β is equal to

A. -1 only

B. 1 only

C. 3 only

D. 3 or 15

Question 9

The angle between the plane $-2x+y+z=8$ and the line $\frac{x+1}{3}=\frac{y-5}{2}=1-z$ is

A. $\sin^{-1}\left(\frac{-\sqrt{21}}{14}\right)$

B. $\sin^{-1}\left(\frac{\sqrt{21}}{14}\right)$

C. $\cos^{-1}\left(\frac{\sqrt{1239}}{42}\right)$

D. $\cos^{-1}\left(\frac{\sqrt{21}}{14}\right)$

Question 10

Consider the differential equation $\frac{dy}{dx}=y\sin(x)$ where $y(x_0)=y_0$. Using Euler's method, with a step size of h, which one of the following is equal y_2?

A. $(y_0+h)(1+h\sin(x_0+h))$

B. $y_0(1+h\sin(x_0))(1+h\sin(x_0+h))$

C. $y_0(1+h\sin(x_0)+h(1+h\sin(x_0+h)))$

D. $y_0e^{(\cos(x_o)-\cos(x+2h))}$

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Question 11

Sand is poured into a conical pile, its volume V changes as its height h and radius r both change with respect to time. Then

A. $\frac{dV}{dt}=\frac{\pi}{3}\left(r^2+2rh\frac{dh}{dt}\right)$

B. $\frac{dV}{dt}=\frac{\pi}{3}\left(2r\frac{dr}{dh}.\frac{dh}{dt}\right)$

C. $\frac{dV}{dt}=\frac{\pi}{3}\left(2rh\frac{dr}{dt}+r^2\frac{dh}{dt}\right)$

D. $\frac{dV}{dt}=\frac{\pi}{3}\left(2r\frac{dr}{dh}+\frac{dh}{dt}\right)$

Question 12

The direction (slope) field for a certain differential equation is shown below.

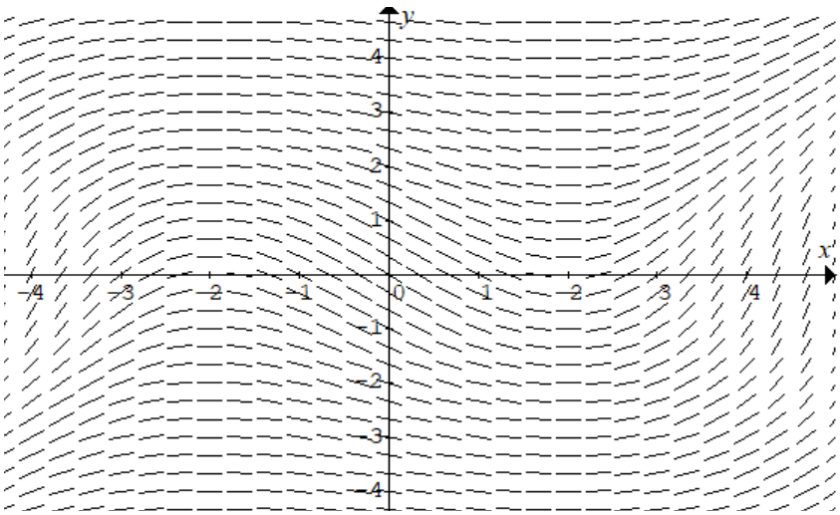
The differential equation could be

A. $\frac{dy}{dx}=\frac{x^2+4}{y+2}$

B. $\frac{dy}{dx}=\frac{x^2+4}{y-2}$

C. $\frac{dy}{dx}=\frac{x^2-4}{y^2+4}$

D. $\frac{dy}{dx}=\frac{x^2-4}{y^2-4}$



Question 13

A particle moves in a straight line. When its displacement from a fixed origin is x metres, its acceleration is given by $\frac{3bx-2b^2}{x^3}$ m/s^2, where b is a non-zero real constant. Given that the particle comes to rest at $x=b$, then it also comes to rest at

A. $x=\frac{b}{2}$

B. $x=2b$

C. $x=\frac{b}{3}$

D. $x=\frac{2b}{3}$

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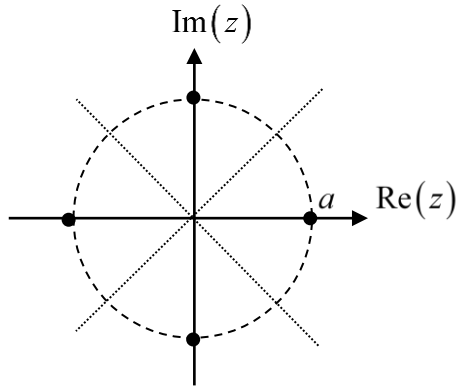
Question 14

Which one of the following diagrams represents the roots of the equation $z^4 + a^4 i = 0$ in the complex plane where a is a positive real constant?

A.



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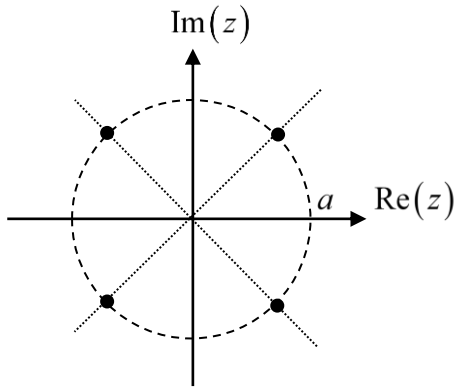
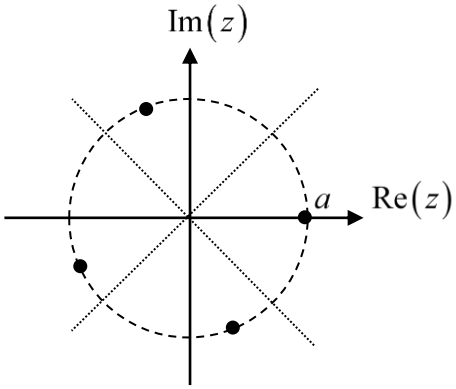
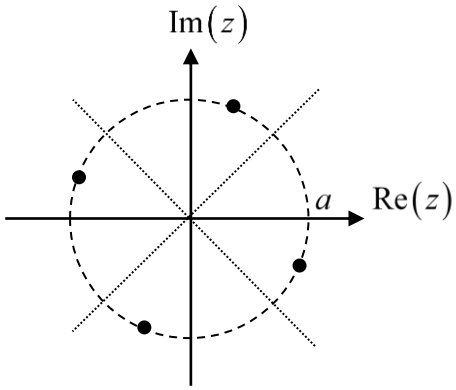


C.



D.



Question 15

A cricketer hits a cricket ball off his toes with an initial velocity of $20\,\text{ms}^{-1}$ at an angle of 40^0. The ball travels in a two dimensional parabolic path and is caught on its downward trajectory at a height of 1.5 m above the ground, by a fieldsman in the outfield. Several students were analyzing the motion of the cricket ball.

Alan stated that the ball reached a maximum height of 8.43 metres, correct to two decimal places above the ground.

Ben stated that the time from when the cricket ball was hit until it was caught in the outfield was 2.5 seconds, correct to one decimal place.

Colin stated that the fieldsman was 38.32 metres, correct to two decimal places from the batsman when the ball was caught in the outfield.

A. Only Alan is correct.

B. Only Ben is correct.

C. Only Colin is correct.

D. All of Alan, Ben and Colin are all correct.

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Question 16

Consider a tank, which initially holds 10 litres of a solution, in which 2 kilograms of salt has been dissolved. Another solution containing 1.5 kilograms of salt per litre is poured into the tank at a rate of $2b$ litres per minute. The well-stirred mixture leaves the tank at a rate of b litres per minutes.

The amount of salt in kilograms in the tank at a time t minutes, Q is given by

$Q(t)=\dfrac{3t^2+60t+40}{2(t+10)}$, then b is equal to

A. 1

B. 2

C. 3

D. 4

Question 17

Two particles A and B are moving, their position vectors at time t are given by

$\underset{\sim}{r}_A(t)=(t^2-4t)\underset{\sim}{i}+(t^2-3t)\underset{\sim}{j}$ and $\underset{\sim}{r}_B(t)=(t^2+2t-15)\underset{\sim}{i}+(t^2-16)\underset{\sim}{j}$ for $t\geq 0$.

Then

A. Both particles travel on part of a parabolic curve.

B. The particles are closest to each when $t=\dfrac{46}{15}$ and the closest distance between them is $\dfrac{17\sqrt{5}}{5}$.

C. The paths of the particles intersect.

D. The two particles collide.

Question 18

A toy car moves, in a straight line, so that at a time t seconds $t\geq 0$, its velocity v ms^{-1} is given by $v(t)=\dfrac{20}{\pi}\arcsin\left(1-\dfrac{t}{4}\right)$. Over the first seven seconds, the distance travelled by the particle in metres is closest to

A. 6.96

B. 22.11

C. 14.53

D. 1266.84

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Question 19

The mean heights of year 12 students at a school was calculated to be 162 cm. A random sample of 36 year 12 students from the school were selected, the mean height of the sample was 160 cm with a standard deviation of 8.5 cm. The minimum confidence level, correct to the nearest whole percent, that could be used to produce a confidence interval that contains μ, based on this sample is

A. 80%

B. 84%

C. 85%

D. 90%

Question 20

A random variable X is normally distributed with a known mean μ and a standard deviation of σ. Consider the distribution of $\bar{X}$ based on repeated random sampling of X. When sampling n_1 times the probability that $\bar{X}$ is less than a certain value is 0.02, and when sampling n_2 times the probability that $\bar{X}$ is greater than the same certain value is 0.998. The ratio $\frac{n_2}{n_1}$ is closest to

A. $\frac{1}{2}$

B. 2

C. 4

D. 10

End of Section A

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Section B

Instructions

- Answer **all** questions in the spaces provided.
- Unless otherwise specified an **exact** answer is required to a question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the **acceleration due to gravity** to have magnitude g m/s^2, where $g = 9.8$.

Question 1 (11 marks)

Let $f(x) = \dfrac{2x^3 - 6x^2 - 20x + 48}{x^3 + x^2 - 10x + 8}$

a.i. State the equations of all the straight line asymptotes on the graph of f.

1 mark

ii. Find the coordinates of all axial intercepts on the graph of f.

1 mark

iii. Explain why the graph of f does not have any turning points.

1 mark

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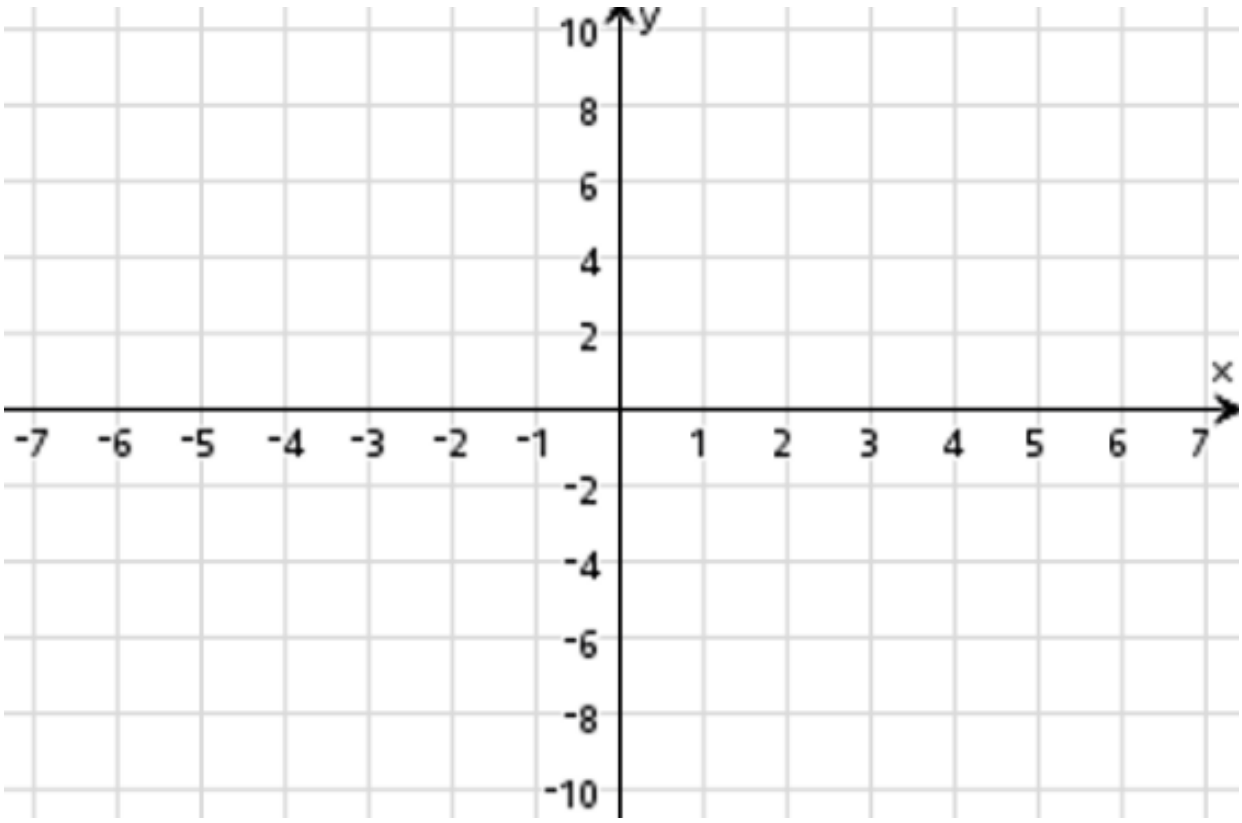
iv. Sketch the graph of f on the set of axes below. Label the asymptotes with their equations, and label the point of inflection with its coordinates, correct to two decimal places.
Label the axial intercepts with their coordinates and any other special features of the graph.

3 marks



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b. Consider now the graph of $g(x)=\dfrac{2x^3-6x^2-20x+48}{x^3-(3+a)x^2+(3a+2)x-2a}$ where $a \in R$.

i. For what value(s) of a does the graph of g have exactly two straight line asymptotes?

1 mark

ii. For what value(s) of a does the graph of g have exactly three straight line asymptotes?

1 mark

iii. For what value(s) of a will the graph of g cross the horizontal asymptote?

1 mark

iv. For what value(s) of a will the graph of g have two turning points?

2 marks

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Question 2 (10 marks)

Let $u = 4 + 2i$

a. **i.** Find the Cartesian equation of the circle $|z - u| = 4$ where $z \in C$.

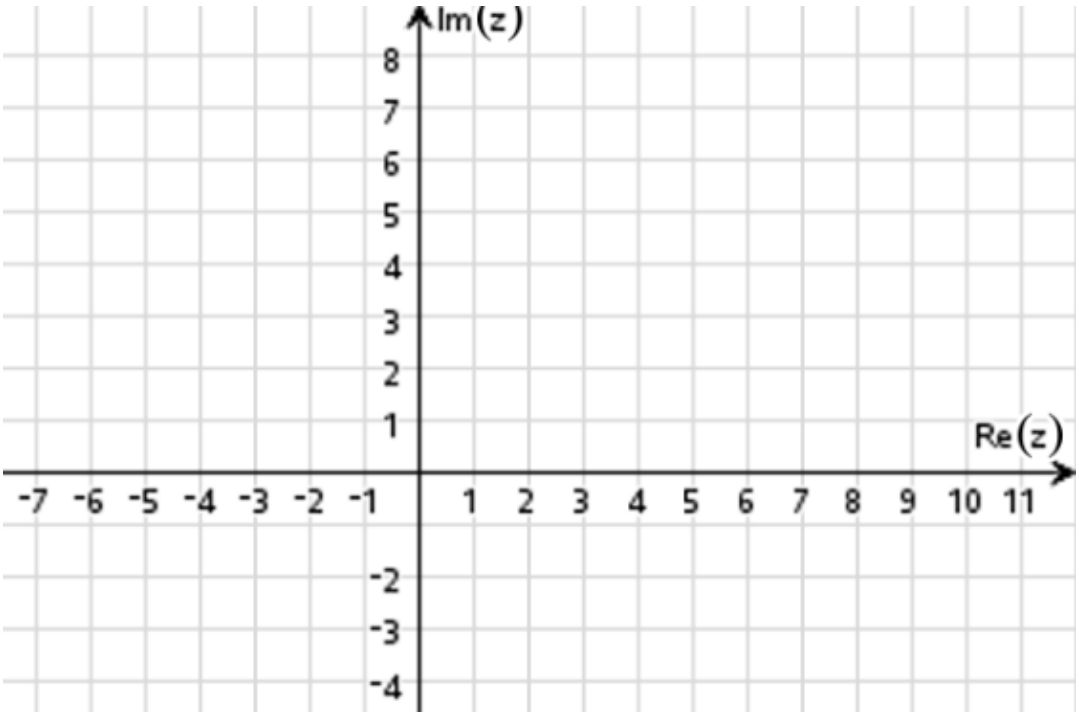
1 mark

ii. The circle intersects the x-axis at two points P and Q where the x coordinates of Q is greater than the x coordinate of P. The circle also intersects the y-axis at the point R. Write down the coordinates of the points P, Q and R.

1 mark

iii. Sketch the circle on the Argand diagram below.

1 mark



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b. **i.** A ray S, which originates at Q also passes through the point R, and can be expressed as $S = \{z : \text{Arg}\left(z - z_Q\right) = k\pi, z \in C\}$. Determine the complex number z_Q and the value of k, and sketch the ray on the diagram in part **a.iii.**

2 marks

ii. The ray divides the circle into a major and minor segment.
Find the area of the minor segment.

2 marks

iii. The ray can also be represented by part of the line $T = \{z \in C : \left|z - z_p\right| = \left|z - z_T\right|\}$, where z_P is the point P. Determine the complex number z_T.

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c. The polynomial $p(z) = z^5 + \alpha z^4 + \beta z^3 + \chi z^2 + \delta z - 200$, where $\alpha, \beta, \chi, \delta \in R$ has both u and $\frac{u}{2}$ amongst its roots. Determine the values of α, β, χ and δ.

1 mark

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Question 3 (11 marks)

A particle moves so that its position vector at a time t is given by

$\underset{\sim}{r}(t) = \cos(t)\underset{\sim}{i} + \cos(3t)\underset{\sim}{j}$ where $0 \le t \le \frac{\pi}{2}$.

a. Express $\cos(3A)$ in terms of $\cos(A)$, and find the Cartesian equation of the curve.

2 marks

b.i. Show $\sin(3A) = 3\sin(A) - 4\sin^3(A)$ and hence show that that the gradient of the curve can be expressed in terms of t in the form $a + b\sin^2(t)$ where $a, b \in Z$.

2 marks

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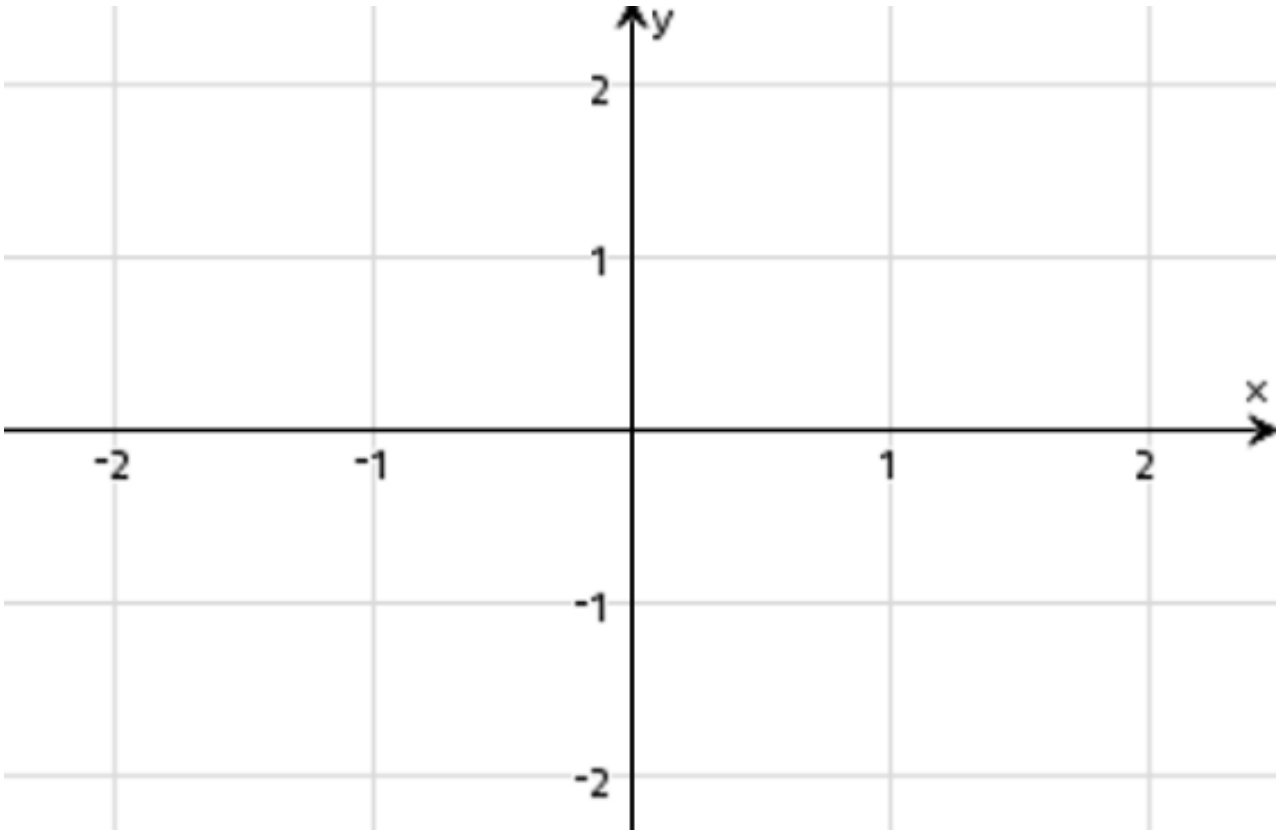
ii. Hence find the coordinate of the turning point on the curve.

1 mark

iii. Sketch the curve on the axes below labelling axial intercepts, any endpoints, and the turning point with their coordinates and show the direction of motion.

2 marks



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d.i. Express the speed of the particle in terms of $\sin(t)$.

2 marks

ii. Find the time when the maximum speed occurs and state the maximum speed, giving both answers correct to three decimal places.

1 mark

e. The total length of the curve can be expressed as the definite integral

$\int_0^{\frac{\pi}{2}} \sin(t)\sqrt{a+b\sin^2(t)+c\sin^4(t)}$ write down the values of a, b and c.

1 mark

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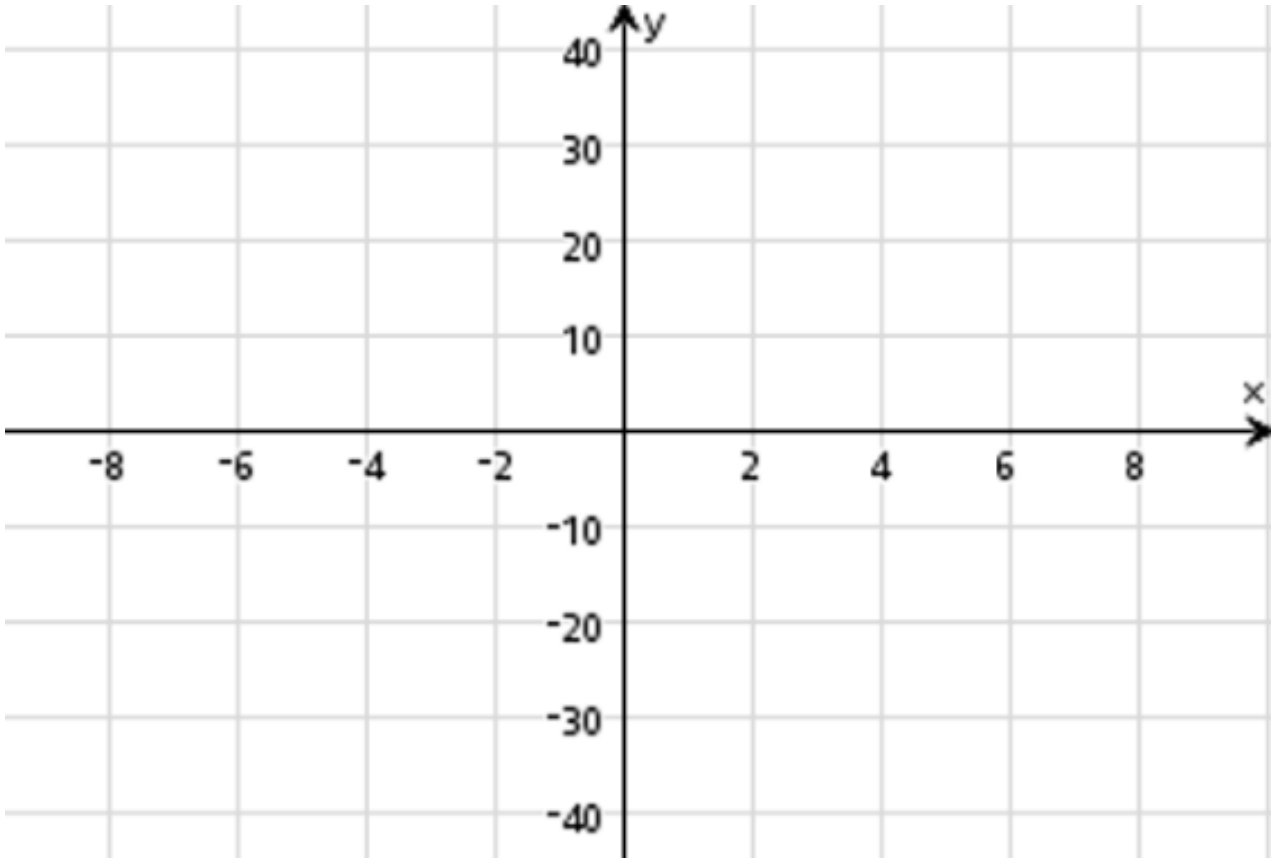
Question 4 (8 marks)

Consider the function f with the rule $f(x) = \frac{80}{\pi}\sin^{-1}\left(\frac{x-4}{5}\right)$.

a. A glass bowl is formed when the part of the curve $y = f(x)$ and the area bounded by the coordinate axes and the line $y = 30$ is rotated about the y-axis. Sketch the graph of the function, and the shape of the bowl on the diagram below.

The dimensions of the bowl are in centimetres.

2 marks



b. Find the diameter of the top of the bowl, giving your answer correct to three decimal places.

1 mark

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c. When the bowl is filled with water to a depth of H centimetres, where $0 < H < 30$, show that the volume V of the water in the bowl in cubic centimetres, is given by the definite integral $V = \pi \int_0^H \left(4 + 5\sin\left(\frac{\pi y}{80}\right)\right)^2 dy$.

1 mark

d. Find a definite integral in terms of y, which gives the total curved surface area of the bowl in square centimetres.

1 mark

e. After the bowl is filled with water to a height of 25 centimetres, a small hole appears in the base and the water leaks out at a rate of $k\sqrt{h}\ \text{cm}^3\,\text{s}^{-1}$, where h is the height of the water remaining in the bowl. If the bowl is empty after 10 **minutes**, determine the value of k, giving your answer correct to three decimal places.

3 marks

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Question 5 (9 marks)

Consider the three points the origin $O(0,0,0)$ $P(3,-2,1)$ and $Q(q,-1,3)$ where $q \in R$.

a. For some particular values of q, the area of the triangle OPQ is $\frac{5\sqrt{6}}{2}$, determine the value(s) of q.

1 mark

b. If the shortest distance from the point Q to the plane $5x+7y-z=8$ is $\frac{\sqrt{3}}{5}$, determine the value(s) of q.

2 marks

c. Find the vector equation in terms of $\underset{\sim}{i}, \underset{\sim}{j}, \underset{\sim}{k}$ and q for the line L_1 passing through the points P and Q.

1 mark

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d. The line L_1, intersects the plane $5x+7y-z=10$, at the point $B(4,by,bz)$, find the value(s) of q , and by and bz.

1 mark

e. Consider another line $L_2 : \underset{\sim}{r}(t)=(2+2t)\underset{\sim}{i}+(-1+t)\underset{\sim}{j}+(3+2t)\underset{\sim}{k}$, If the two lines L_1 and L_2 are parallel find the value of q and the shortest distance between the two lines L_1 and L_2 .

2 marks

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f. If the shortest distance from the line L_1 to the point $A(-1,2,-2)$ is $\sqrt{35}$, determine the value(s) of q.

2 marks

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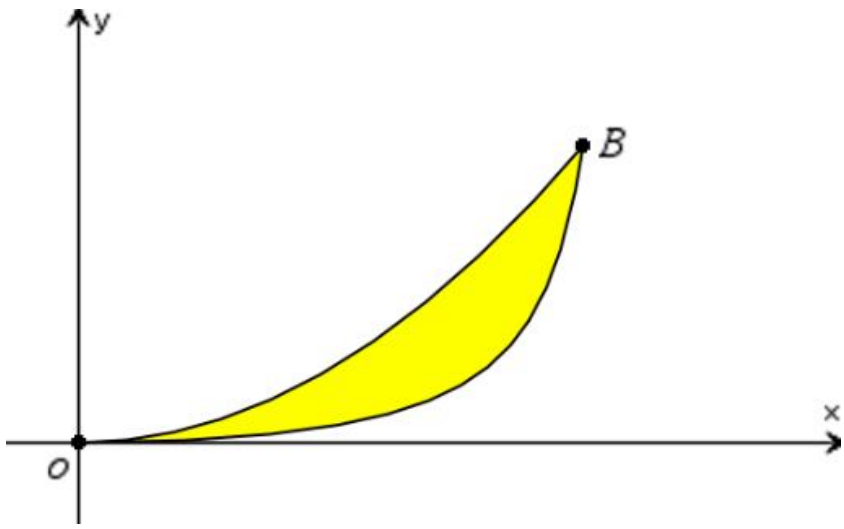
Question 6 (11 marks)

a. The diagram below shows part of the graphs of two functions, $f:[0,\infty)\to R,\ f(x)=\frac{x^2}{10}$ and $g:[0,\infty)\to R,\ g(x)=\sec\left(\frac{3x}{16}\right)-1$. The two graphs intersect at the point B.



i. Determine the coordinates of the point B, giving your answer correct to two decimal places.

1 mark

ii. When the area bounded by the graphs the origin O and the point B, is rotated about the x-axis, it forms the shape of a banana. If dimensions for the x and y axes are in centimetres, Determine the volume of the banana, giving your answer in cubic centimetres correct to three decimal places.

1 mark

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b. The weights of medium sized bananas are normally distributed with a mean of 118 grams with a standard deviation of 6 grams.

A bunch of bananas contains n bananas. Determine the value of n, if the probability that the bunch of bananas weighs less than 1.5 kg is 0.058.

1 mark

c. A farm grows bananas, the owner of the farm is not sure if his bananas are medium sized or not. He carefully weighs and measures the weights of 30 bananas and finds their mean weight to be 116 grams with a standard deviation of 3 grams. The owner will carry out a two-sided hypothesis test.

i. Write down the null and alternative hypothesis.

1 mark

ii. Determine the p-value, giving your answer correct to three decimal places and with reasons decide if he accepts the null hypothesis, at the 5% level of significance.

2 marks

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d.i. If in fact the mean weights of medium sized bananas is assumed to be 116.5 grams, and not 118 grams, determine the value of the type II error, using a sample of thirty bananas at the 5% level of significance. Assume that the population standard deviation of medium sized bananas is 6 grams and give your answer correct to three decimal places.

2 marks

ii. If in fact the mean weights of medium sized bananas is assumed to be 116.5 grams, and not 118 grams, using a sample size of n bananas at the 5% level of significance, find the minimum sample size required so that the probability of a type II error is less than 5%.? Assume that the population standard deviation of medium sized bananas is still 6 grams

2 marks

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e. The weight of the peel (skin) of a medium sized banana is 15 grams, with a standard deviation of 0.75 grams. The edible part of a banana is the peeled part without the skin. Jack peels and eats two medium sized bananas, determine the probability the amount of banana that Jack ate exceeds 200 grams, give your answer correct to four decimal places.

1 mark

End of Section B

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PO Box 3229
Cotham Vic 3101
Australia

PayID: 47065111373
Email: kilbaha@gmail.com
Tel: +61 413 425 374
Web: https://kilbaha.com.au

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Specialist Mathematics
Examination 2
2025 Formula Sheet

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Mensuration

area of a circle segment	$\frac{r^2}{2}(\theta - \sin(\theta))$	volume of a sphere	$\frac{4}{3}\pi r^3$
volume of a cylinder	$\pi r^2 h$	area of a triangle	$\frac{1}{2}bc\sin(A)$
volume of a cone	$\frac{1}{3}\pi r^2 h$	sine rule	$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$
volume of a pyramid	$\frac{1}{3}Ah$	cosine rule	$c^2 = a^2 + b^2 - 2ab\cos(C)$

Algebra, number and structure (complex numbers)

$z = x + yi = r(\cos(\theta) + i\sin(\theta)) = r\,\text{cis}(\theta)$	$\lvert z\rvert = \sqrt{x^2 + y^2} = r$	
$-\pi < \text{Arg}(z) \le \pi$	$z_1 z_2 = r_1 r_2\,\text{cis}(\theta_1 + \theta_2)$	
$\frac{z_1}{z_2} = \frac{r_1}{r_2}\,\text{cis}(\theta_1 - \theta_2)$	de Moivre's theorem	$z^n = r^n\text{cis}(n\theta)$

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Data analysis, probability and statistics

for independent random variables $X_1, X_2 .... X_n$	$\mathrm{E}(aX_1+b)=a\mathrm{E}(X_1)+b$ $\mathrm{E}(a_1X_1+a_2X_2+...+a_nX_n)$ $=a_1\mathrm{E}(X_1)+a_2\mathrm{E}(X_2)+...+a_n\mathrm{E}(X_n)$	
	$\mathrm{Var}(aX_1+b)=a^2\mathrm{Var}(X_1)$ $\mathrm{Var}(a_1X_1+a_2X_2+...a_nX_n)$ $=a_1^2\mathrm{Var}(X_1)+a_2^2\mathrm{Var}(X_2)+...+a_n^2\mathrm{Var}(X_n)$	
for independent identically distributed variables $X_1, X_2 .... X_n$	$\mathrm{E}(X_1+X_2+...+X_n)=n\mu$	
	$\mathrm{Var}(X_1+X_2+...+X_n)=n\sigma^2$	
approximate confidence interval for μ	$\left(\bar{x}-z\frac{s}{\sqrt{n}}, \bar{x}+z\frac{s}{\sqrt{n}}\right)$	
distribution of sample mean $\bar{X}$	mean	$\mathrm{E}(\bar{X})=\mu$
	variance	$\mathrm{Var}(\bar{X})=\frac{\sigma^2}{n}$

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Calculus

$\frac{d}{dx}\left(x^n\right)=nx^{n-1}$	$\int x^n dx=\frac{1}{n+1}x^{n+1}+c\ ,\ n\neq -1$
$\frac{d}{dx}\left(e^{ax}\right)=ae^{ax}$	$\int e^{ax}dx=\frac{1}{a}e^{ax}+c$
$\frac{d}{dx}\left(\log_e(x)\right)=\frac{1}{x}$	$\int \frac{1}{x}dx=\log_e\left(\lvert x\rvert\right)+c$
$\frac{d}{dx}\left(\sin(ax)\right)=a\cos(ax)$	$\int \sin(ax)dx=-\frac{1}{a}\cos(ax)+c$
$\frac{d}{dx}\left(\cos(ax)\right)=-a\sin(ax)$	$\int \cos(ax)dx=\frac{1}{a}\sin(ax)+c$
$\frac{d}{dx}\left(\tan(ax)\right)=a\sec^2(ax)$	$\int \sec^2(ax)dx=\frac{1}{a}\tan(ax)+c$
$\frac{d}{dx}\left(\cot(ax)\right)=-a\,\mathrm{cosec}^2(ax)$	$\int \mathrm{cosec}^2(ax)dx=-\frac{1}{a}\cot(ax)+c$
$\frac{d}{dx}\left(\sec(ax)\right)=a\sec(ax)\tan(ax)$	$\int \sec(ax)\tan(ax)dx=\frac{1}{a}\sec(ax)+c$
$\frac{d}{dx}\left(\mathrm{cosec}(ax)\right)=-a\,\mathrm{cosec}(ax)\cot(ax)$	$\int \mathrm{cosec}(ax)\cot(ax)dx=-\frac{1}{a}\mathrm{cosec}(ax)+c$
$\frac{d}{dx}\left(\sin^{-1}(ax)\right)=\frac{a}{\sqrt{1-(ax)^2}}$	$\int \frac{1}{\sqrt{a^2-x^2}}dx=\sin^{-1}\left(\frac{x}{a}\right)+c,\ a>0$
$\frac{d}{dx}\left(\cos^{-1}(ax)\right)=\frac{-a}{\sqrt{1-(ax)^2}}$	$\int \frac{-1}{\sqrt{a^2-x^2}}dx=\cos^{-1}\left(\frac{x}{a}\right)+c,\ a>0$
$\frac{d}{dx}\left(\tan^{-1}(ax)\right)=\frac{a}{1+(ax)^2}$	$\int \frac{a}{a^2+x^2}dx=\tan^{-1}\left(\frac{x}{a}\right)+c$
	$\int (ax+b)^n dx=\frac{1}{a(n+1)}(ax+b)^{n+1}+c,\ n\neq -1$
	$\int (ax+b)^{-1}dx=\frac{1}{a}\log_e\left(\lvert ax+b\rvert\right)+c$

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Calculus- continued

product rule	$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
quotient rule	$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
chain rule	$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$
integration by parts	$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$
Euler's method	If $\frac{dy}{dx} = f(x),\ x_0 = a$ and $y_0 = b$, then $x_{n+1} = x_n + h$ and $y_{n+1} = y_n + hf(x_n, y_n)$
arc length parametric	$\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$
surface area Cartesian about the *x*-axis	$\int_{x_1}^{x_2} 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$
surface area Cartesian about the *y*-axis	$\int_{y_1}^{y_2} 2\pi x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$
surface area parametric about the *x*-axis	$\int_{t_1}^{t_2} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$
surface area parametric about the *y*-axis	$\int_{t_1}^{t_2} 2\pi x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

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Kinematics

acceleration	$a=\frac{d^2x}{dt^2}=\frac{dv}{dt}=v\frac{dv}{dx}=\frac{d}{dx}\left(\frac{1}{2}v^2\right)$	
constant acceleration formulas	$v=u+at$	$s=ut+\frac{1}{2}t^2$
	$v^2=u^2+2as$	$s=\frac{1}{2}(u+v)t$

Vectors in two and three dimensions

$\underset{\sim}{r}(t)=x(t)\underset{\sim}{i}+y(t)\underset{\sim}{j}+z(t)\underset{\sim}{k}$	$\lvert\underset{\sim}{r}(t)\rvert=\sqrt{x(t)^2+y(t)^2+z(t)^2}$
	$\dot{\underset{\sim}{r}}(t)=\frac{d\underset{\sim}{r}}{dt}=\frac{dx}{dt}\underset{\sim}{i}+\frac{dy}{dt}\underset{\sim}{j}+\frac{dz}{dt}\underset{\sim}{k}$
for $\underset{\sim}{r}_1=x_1\underset{\sim}{i}+y_1\underset{\sim}{j}+z_1\underset{\sim}{k}$ and $\underset{\sim}{r}_2=x_2\underset{\sim}{i}+y_2\underset{\sim}{j}+z_2\underset{\sim}{k}$	vector scalar product $\underset{\sim}{r}_1\cdot\underset{\sim}{r}_2=\lvert r_1\rvert\lvert r_2\rvert\cos(\theta)=x_1x_2+y_1y_2+z_1z_2$
	vector cross product $\underset{\sim}{r}_1\times\underset{\sim}{r}_2=\begin{vmatrix}\underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k}\\ x_1 & y_1 & z_1\\ x_2 & y_2 & z_2\end{vmatrix}=(y_1z_2-y_2z_1)\underset{\sim}{i}+(x_2z_1-x_1z_2)\underset{\sim}{j}+(x_1y_2-x_2y_1)\underset{\sim}{k}$
vector equation of a line	$\underset{\sim}{r}(t)=\underset{\sim}{r}_1+t\underset{\sim}{r}_2=(x_1+x_2t)\underset{\sim}{i}+(y_1+y_2t)\underset{\sim}{j}+(z_1+z_2t)\underset{\sim}{k}$
parametric equation of line	$x(t)=x_1+x_2t \quad y(t)=y_1+y_2t \quad z(t)=z_1+z_2t$
vector equation of a plane	$\underset{\sim}{r}(s,t)=\underset{\sim}{r}_0+s\underset{\sim}{r}_1+t\underset{\sim}{r}_2$ $=(x_0+x_1s+x_2t)\underset{\sim}{i}+(y_0+y_1s+y_2t)\underset{\sim}{j}+(z_0+z_1s+z_2t)\underset{\sim}{k}$
parametric equation of a plane	$x(s,t)=x_0+x_1s+x_2t \quad y(s,t)=y_0+y_1s+y_2t \quad z(s,t)=z_0+z_1s+z_2t$
Cartesian equation of a plane	$ax+by+cz=d$

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Circular (trigonometric) functions

$\cos^2(x)+\sin^2(x)=1$	
$1+\tan^2(x)=\sec^2(x)$	$\cot^2(x)+1=\operatorname{cosec}^2(x)$
$\sin(x+y)=\sin(x)\cos(y)+\cos(x)\sin(y)$	$\sin(x-y)=\sin(x)\cos(y)-\cos(x)\sin(y)$
$\cos(x+y)=\cos(x)\cos(y)-\sin(x)\sin(y)$	$\cos(x-y)=\cos(x)\cos(y)+\sin(x)\sin(y)$
$\tan(x+y)=\dfrac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}$	$\tan(x-y)=\dfrac{\tan(x)-\tan(y)}{1+\tan(x)\tan(y)}$
$\sin(2x)=2\sin(x)\cos(x)$	
$\cos(2x)=\cos^2(x)-\sin^2(x)$ $=2\cos^2(x)-1=1-2\sin^2(x)$	$\tan(2x)=\dfrac{2\tan(x)}{1-\tan^2(x)}$
$\sin^2(ax)=\dfrac{1}{2}(1-\cos(2ax))$	$\cos^2(ax)=\dfrac{1}{2}(1+\cos(2ax))$

End of Formula Sheet

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PO Box 3229
Cotham Vic 3101
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